

VASUDEV NAIR P

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PROFESSIONAL SUMMARY

Final-year Computer Science student (S7, graduating June 2027) with hands-on experience in full-stack development, backend systems, and AI/LLM applications. Co-founded a cloud workstation platform and built production-grade software across multiple domains. National-level ideathon winner with expertise in building scalable systems that solve real problems.

TECHNICAL SKILLS

Backend: Node.js, Express.js, FastAPI, Python

Databases: PostgreSQL, MySQL, Firebase, data modeling, query optimization

AI / LLM: Groq API, Gemini API, prompt engineering, NLP concepts

Frontend / Mobile: React, HTML5, CSS3, Kotlin (Android)

Cloud & DevOps: AWS, DigitalOcean, Heroku, deployment automation

Architecture & Tools: REST API design, WebSockets, real-time systems, blockchain, system design

Development Tools: Git, version control, Python, Java, C, JavaScript

PROFESSIONAL EXPERIENCE

Co-Founder & Chief Operating Officer — Virga Cloud Workstation Platform (2021 – 2025)

virga.tech

- Conceptualized and launched a cloud workstation platform for GPU-intensive applications including game development, 3D rendering, and content creation
- Built the MVP using cloud infrastructure and containerization for on-demand compute resource provisioning
- Led a cross-functional team, managed the technical roadmap, and drove strategic market decisions
- Gained end-to-end experience in product lifecycle management, infrastructure architecture, and cloud service integration

Tech Stack: Kotlin Android, Node.js/Express, PostgreSQL, WebSockets

KEY PROJECTS

Intelligent Travel Companion Application (S6 Capstone Project)

- Built a unified multi-modal travel platform integrating bus, metro, and ferry routes across Kerala into a single journey planning interface
- Implemented a real-time backend with live vehicle tracking and route optimization algorithms
- Engineered a sun-position-based seating recommendation system using astronomical calculations
- Developed a GPS proximity alarm with background location tracking and one-tap emergency SOS feature

Tech Stack: Python, Groq API, Gemini API, FastAPI, LLM orchestration

LLM-Based Intelligent Systems (Multiple Personal Projects)

- Built several LLM-powered applications leveraging free-tier Groq and Gemini APIs for cost-effective deployment
- Implemented conversational systems with context management, prompt optimization, and effective API integration patterns
- Deployed to cloud infrastructure (AWS, Heroku, DigitalOcean) with attention to API rate limiting and cost optimization

Tech Stack: Python, FastAPI, LLM APIs, cloud deployment

Blockchain-Based Credential System (SIH 2024–25)

- Designed and implemented a secure certificate issuance and verification system using blockchain for immutability and transparency
- Developed certificate generation logic with cryptographic hashing ensuring tamper-proof credential tracking
- College-level SIH competition winner, demonstrating ability to solve real-world problems with emerging technologies

Tech Stack: Python, blockchain integration, cryptographic verification

COMPETITIONS & ACHIEVEMENTS

- **National-Level Ideathon Winner** — Won a national-level ideathon, demonstrating problem-solving and innovative thinking at a competitive scale
- **College-Level SIH 2024–25 Winner** — Won the Smart India Hackathon at college level, executing a complex technical solution under time constraints
- **Multiple College Ideation Wins** — Consistently recognized for conceptual thinking and identifying market opportunities
- **Hackathon Participant** — CODE-OF-DUTY (AI-based food waste optimization) and MITS Eneriya 2025, building full-stack prototypes and MVPs

EDUCATION

B.Tech in Computer Science and Engineering

St. Joseph's College of Engineering and Technology, Palai, Kerala

Entering 7th Semester | Expected Graduation: June 2027

TECHNICAL CAPABILITIES & INTERESTS

- **Backend Development:** Strong proficiency in building scalable REST APIs, real-time systems, database design, and server-side architecture
- **Cloud Deployment:** Experienced deploying and managing applications on AWS, Heroku, and DigitalOcean with infrastructure and cost optimization expertise
- **AI & Intelligent Systems:** Hands-on experience with Large Language Model APIs, prompt engineering, and building AI-powered applications
- **Full-Stack Capability:** Comfortable working across the entire stack — from database design through backend systems to frontend interfaces
- **Leadership & Entrepreneurship:** Co-founder and COO experience managing team dynamics, strategic decisions, and product lifecycle from idea to market
- **Problem-Solving:** Passionate about building scalable systems with meaningful real-world impact, with particular interest in data systems and real-time processing